

Mathematics for AI

Linear Transformations and Matrix Representations

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Learning Outcomes

By the end of this lecture, you should be able to:

- Understand the concept of linear transformations
- Represent linear transformations using matrices
- Explore the connection between linear maps and matrix multiplication
- Apply linear transformations to vectors
- Recognize examples from AI

Linear Transformation: Definition

- A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation if for all vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and scalars $c \in \mathbb{R}$:
 - Additivity: $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
 - Homogeneity: $T(c\mathbf{u}) = cT(\mathbf{u})$
- These properties ensure that the transformation preserves the structure of the vector space
- Linear transformations map the origin to the origin and preserve lines and parallelism

Examples of Linear Transformations

- Identity transformation: $T(\mathbf{x}) = \mathbf{x}$ (leaves vector unchanged)
- Scaling: $T(\mathbf{x}) = k\mathbf{x}$, scales vector length by k
- Rotation in 2D: rotates a vector around the origin by a given angle
- Reflection: e.g., over x-axis or y-axis
- Projection: maps vectors onto a subspace (e.g., x-axis)
- Shearing: distorts shape by shifting one coordinate proportionally to another, while preserving area in some cases (e.g., horizontal shear)
- All these transformations can be represented using matrices

Matrix Representation of Linear Transformations

- Any linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be written as:

$$T(\mathbf{x}) = A \mathbf{x}$$

- Here:
 - $\mathbf{x}$ is an $n \times 1$ vector
 - A is an $m \times n$ matrix
 - $T(\mathbf{x})$ is an $m \times 1$ vector
- Intuition: A encodes exactly how T stretches/rotates/projects each coordinate direction.

Constructing the Matrix of a Transformation

- Standard basis of $\mathbb{R}^n$: $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$

- Apply T to basis vectors:

$$T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$$

- Form A by columns:

$$A = [T(\mathbf{e}_1) \ T(\mathbf{e}_2) \ \cdots \ T(\mathbf{e}_n)]$$

- For any $\mathbf{x} = (x_1, \dots, x_n)$:

$$T(\mathbf{x}) = \sum_{i=1}^n x_i T(\mathbf{e}_i)$$

- Takeaway: The i -th column of A is $T(\mathbf{e}_i)$.

Constructing the Matrix of a Transformation

Examples

Rotation by 90° in $\mathbb{R}^2$

- Basis: $\mathbf{e}_1 = (1, 0), \mathbf{e}_2 = (0, 1)$
- Transformation: $T(x, y) = (-y, x)$
- Images: $T(\mathbf{e}_1) = (0, 1), T(\mathbf{e}_2) = (-1, 0)$
- Matrix: $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

Shear (x by y)

- Basis: $\mathbf{e}_1 = (1, 0), \mathbf{e}_2 = (0, 1)$
- Transformation: $T(x, y) = (x + y, y)$
- Images: $T(\mathbf{e}_1) = (1, 0), T(\mathbf{e}_2) = (1, 1)$
- Matrix: $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Projection onto the x-axis

- Basis: $\mathbf{e}_1 = (1, 0), \mathbf{e}_2 = (0, 1)$
- Transformation: $T(x, y) = (x, 0)$
- Images: $T(\mathbf{e}_1) = (1, 0), T(\mathbf{e}_2) = (0, 0)$
- Matrix: $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

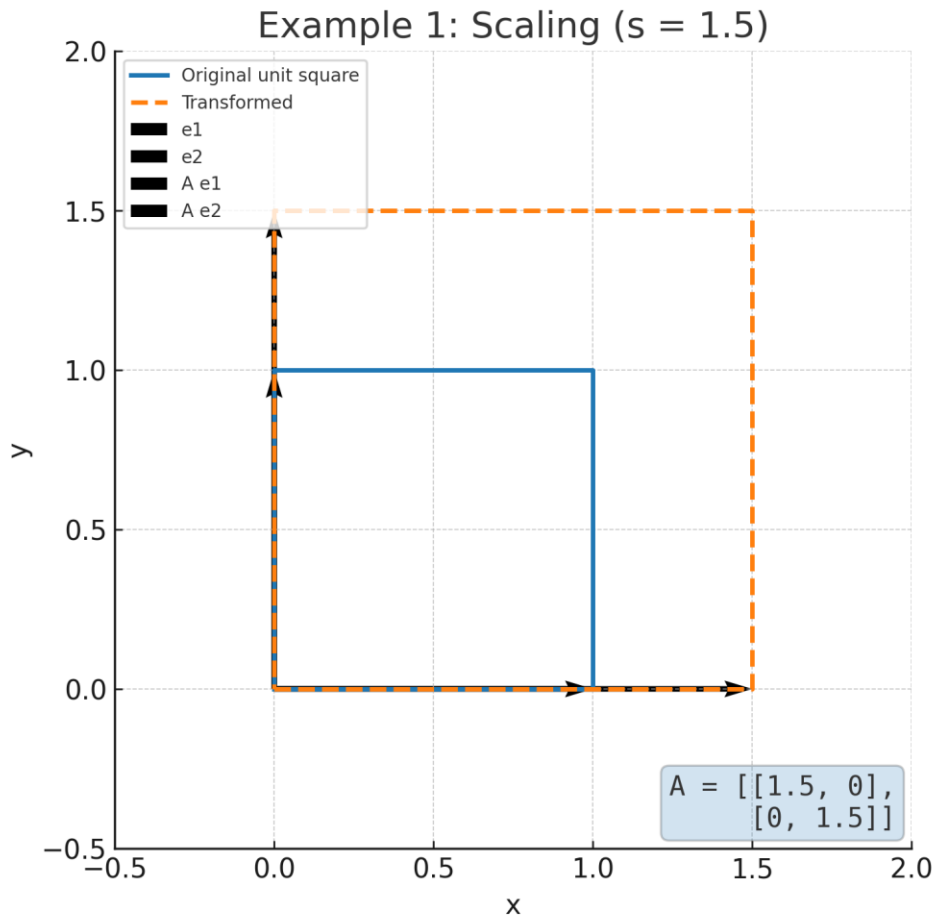
Reflection across the x-axis

- Basis: $\mathbf{e}_1 = (1, 0), \mathbf{e}_2 = (0, 1)$
- Transformation: $T(x, y) = (x, -y)$
- Images: $T(\mathbf{e}_1) = (1, 0), T(\mathbf{e}_2) = (0, -1)$
- Matrix: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Geometric Interpretation

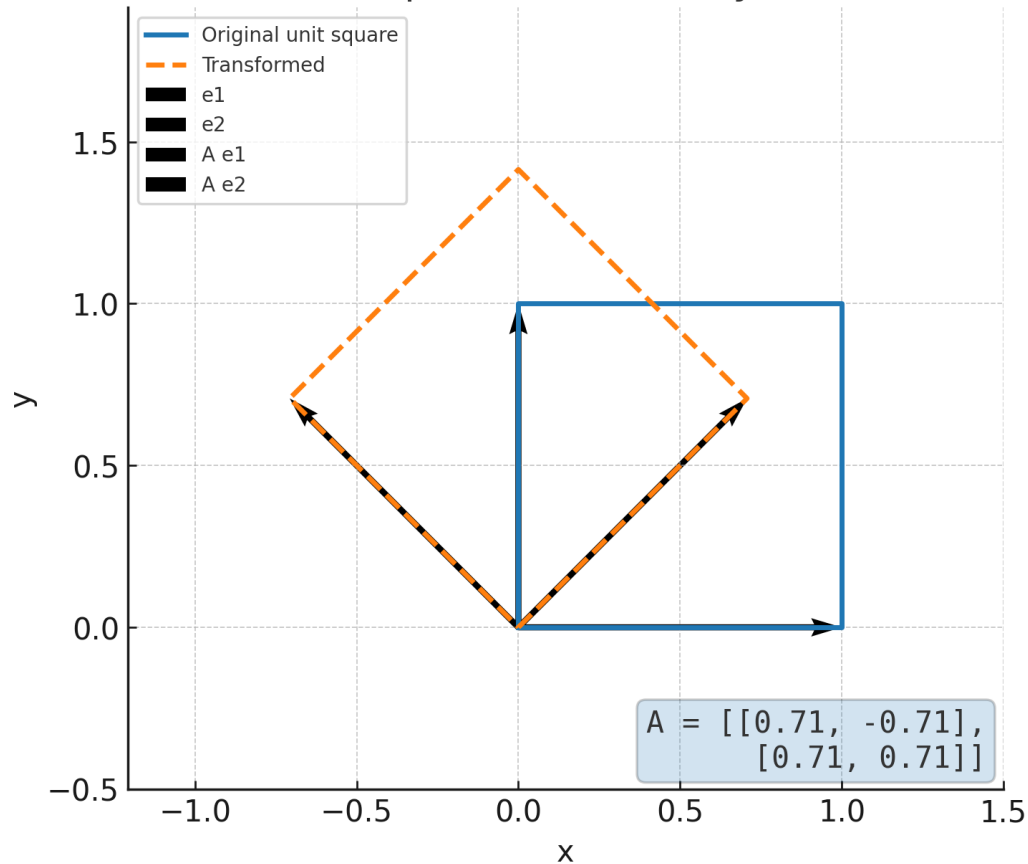
- Linear transformations can stretch, rotate, reflect, shear, or project vectors
- Grid diagrams help visualize the effect on the entire space:
 - **Example 1: Scaling** — The unit square becomes a larger or smaller square depending on the scalar factor.
 - **Example 2: Rotation** — Rotating the unit vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ by 45° transforms the square into a rotated square.
 - **Example 3: Shear** — A shear along the x-axis skews the unit square into a parallelogram with the same base height.
 - **Example 4: Projection** — Projecting onto the x-axis flattens all y-components to 0, collapsing the unit square into a line segment.
 - **Example 5: Area Scaling** — The matrix $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ scales the unit square into a rectangle of area 6. The area scaling factor is given by $\det(A) = 6$.
- Determinant $\det(A)$ provides area scaling factor in 2D

Geometric Interpretation

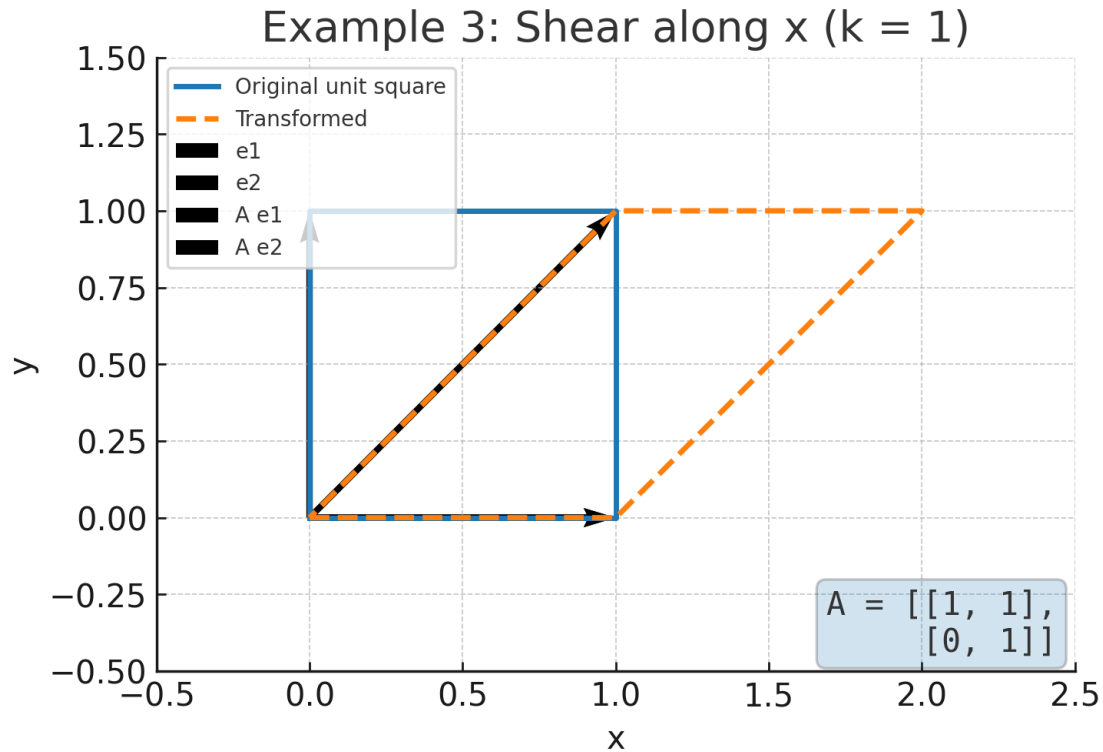


Geometric Interpretation

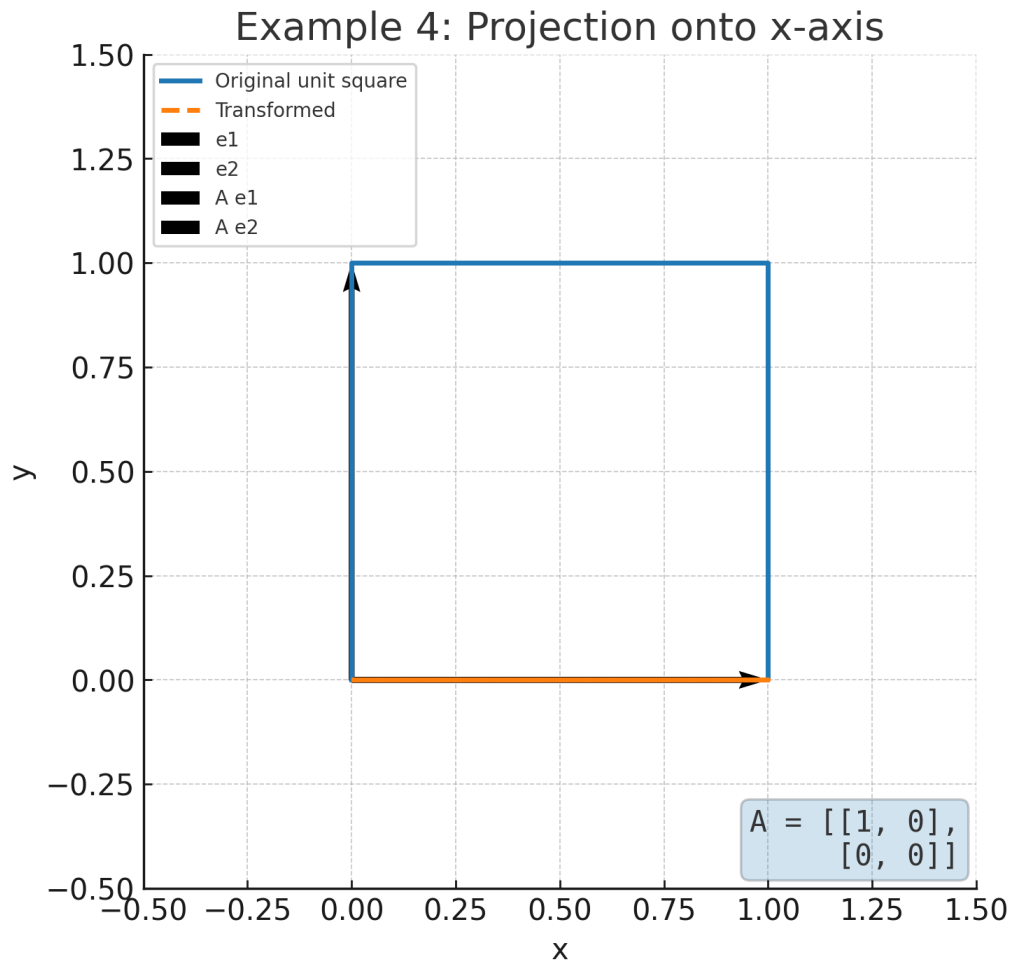
Example 2: Rotation by 45°



Geometric Interpretation

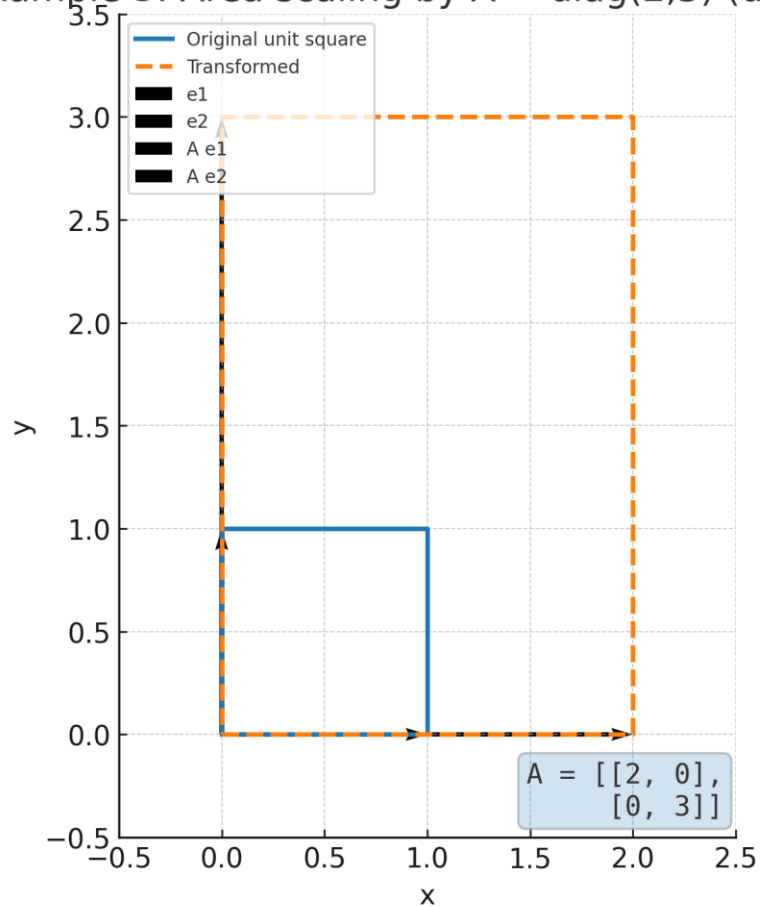


Geometric Interpretation



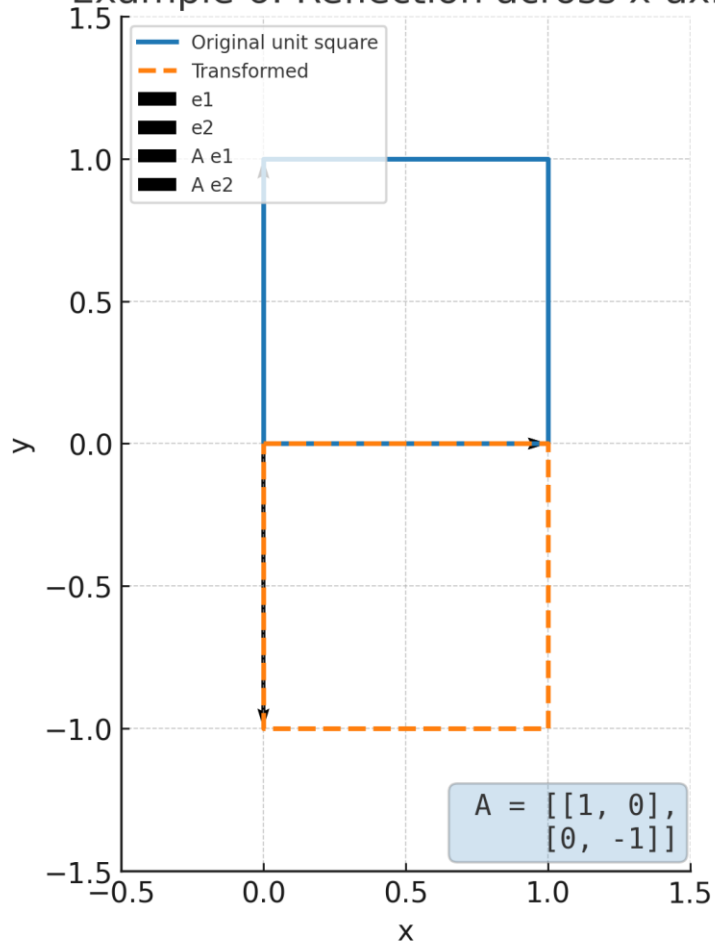
Geometric Interpretation

Example 5: Area scaling by $A = \text{diag}(2,3)$ ($\det = 6$)



Geometric Interpretation

Example 6: Reflection across x-axis



Common 2D Transformations (Matrices)

- Rotation by angle θ : $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
- Reflection across x-axis: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
- Reflection across y-axis: $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$
- Reflection across xy-plane (3D): $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$
- Projection onto x-axis: $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
- Shear along x-axis ($x \leftarrow x + ky$): $A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$
- Shear along y-axis ($y \leftarrow y + kx$): $A = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$
- Shear along z-axis (3D) ($z \leftarrow z + kx$): $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ k & 0 & 1 \end{bmatrix}$

Linear Transformations in AI

Neural network layer as a linear transformation

What it is (affine map).

A dense layer applies

$$T(x) = Wx + b, \quad x \in \mathbb{R}^d, \quad W \in \mathbb{R}^{m \times d}, \quad b \in \mathbb{R}^m.$$

- **Linear part:** Wx (stretches/rotates/skews the input space).
- **Bias:** b (shifts everything; makes it affine, not strictly linear).

Geometric view.

- Columns of W are the images of the basis vectors: $Ae_1, \dots, Ae_d$.
- $\det(W)$ (when $m = d$) = area/volume scaling factor; sign indicates orientation flip.

Linear Transformations in AI

Neural network layer as a linear transformation

Worked mini-example (2→2 layer).

$$W = \begin{bmatrix} 2 & 1 \\ 0 & \frac{1}{2} \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

1. Linear map:

$$Wx = \begin{bmatrix} 2 \cdot 3 + 1 \cdot 2 \\ 0 \cdot 3 + \frac{1}{2} \cdot 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \end{bmatrix}$$

2. Add bias (shift):

$$T(x) = Wx + b = \begin{bmatrix} 8 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \end{bmatrix}$$

Takeaways.

- Wx alone is a pure linear transformation; adding b translates the result.
- In a full NN, nonlinearity $\sigma(\cdot)$ is applied next: $y = \sigma(Wx + b)$.
- Shape check (common source of bugs): $x : [d] \rightarrow Wx : [m]$.

Linear Transformations in AI

Data augmentation as linear transforms on image coordinates

Act on pixel coordinates (u, v) with a 2×2 matrix A (linear) and optionally a translation t (affine):

$$\begin{bmatrix} u' \\ v' \end{bmatrix} = A \begin{bmatrix} u \\ v \end{bmatrix} + t.$$

(Use homogeneous coordinates for a single 3×3 matrix if translations are included.)

Common 2D transforms (matrices A).

- Rotation by θ :

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

- Scaling: $S(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$

- Shear (x): $Sh_x(k) = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$

- Reflection across x-axis: $Ref_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

- Projection to x-axis (flatten y): $P_x = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Linear Transformations in AI

Data augmentation as linear transforms on image coordinates

Worked mini-example (compose rotation + scaling).

Rotate by 30° then scale $s_x = 1.2$, $s_y = 0.8$: $A = S R$.

With $\cos 30^\circ = \sqrt{3}/2 \approx 0.866$, $\sin 30^\circ = 0.5$,

$$R = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix}, \quad S = \begin{bmatrix} 1.2 & 0 \\ 0 & 0.8 \end{bmatrix}, \quad A = \begin{bmatrix} 1.039 & -0.6 \\ 0.4 & 0.693 \end{bmatrix}.$$

Apply to pixel $(u, v) = (100, 50)$:

$$\begin{bmatrix} u' \\ v' \end{bmatrix} = A \begin{bmatrix} 100 \\ 50 \end{bmatrix} = \begin{bmatrix} 1.039 \cdot 100 - 0.6 \cdot 50 \\ 0.4 \cdot 100 + 0.693 \cdot 50 \end{bmatrix} = \begin{bmatrix} 79.0 \\ 73.7 \end{bmatrix} \text{ (approx).}$$

Area effect: $\det(A) = \det(S) \det(R) = (1.2 \cdot 0.8) \cdot 1 = 0.96 \Rightarrow$ slight area shrink (helps reason about scale of objects).

Linear Transformations in AI

Feature Reduction via PCA

Idea

- PCA rotates data to new axes of maximum variance.
- Transformation is linear:

$$Z = \tilde{X}U_k$$

where $\tilde{X}$ = centered data, U_k = top k eigenvectors, Z = projected data.

Example (2D $\rightarrow$ 1D)

- Data: $(2, 2), (3, 3), (-2, -2), (-3, -3)$
- Eigenvectors:
 $u_1 = \frac{1}{\sqrt{2}}(1, 1), \lambda_1 = 13$
 $u_2 = \frac{1}{\sqrt{2}}(-1, 1), \lambda_2 = 0$
- Projection:

$$z = (3, 3) \cdot u_1 = \frac{6}{\sqrt{2}} \approx 4.24$$

- Reconstruction:

$$\hat{x} = zu_1 \approx (3, 3)$$

Linear Transformations in AI Word Embedding

Idea: Words are first represented as one-hot vectors, then mapped through a linear transformation (the embedding matrix) into a dense vector space that captures meaning.

- One-hot encoding (for indexing words):

$$e_{\text{cat}} = [1, 0, 0]^T, \quad e_{\text{dog}} = [0, 1, 0]^T, \quad e_{\text{car}} = [0, 0, 1]^T$$

- Embedding matrix (learned parameters):

Row 1 = embedding for "cat"

Row 2 = embedding for "dog"

Row 3 = embedding for "car"

$$E = \begin{bmatrix} 1.0 & 0.1 \\ 0.9 & 0.0 \\ 0.1 & 0.9 \end{bmatrix}, \quad E \in \mathbb{R}^{V \times d}$$

- Linear transformation:

$$\text{embed}(\text{dog}) = E^T e_{\text{dog}} = \begin{bmatrix} 1.0 & 0.9 & 0.1 \\ 0.1 & 0.0 & 0.9 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.9 \\ 0.0 \end{bmatrix}$$

- Example (dog):

$$E^T e_{\text{dog}} = \begin{bmatrix} 0.9 \\ 0.0 \end{bmatrix}$$

- Outcome: one-hot input $\rightarrow$ dense embedding vector in $\mathbb{R}^d$
 - "dog" $\rightarrow [0.9, 0.0]^T$
 - Captures semantic similarity between words

Composition of Linear Transformations

- Let $T_1(\mathbf{x}) = A\mathbf{x}$, $T_2(\mathbf{x}) = B\mathbf{x}$
- Then the composition $T = T_2 \circ T_1$ is: $T(\mathbf{x}) = B(A\mathbf{x}) = (BA)\mathbf{x}$
- Matrix multiplication corresponds to function composition
- Order matters: $BA \neq AB$ in general

Composition of Linear Transformations Example

Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix} \quad (\text{scale } x \text{ by } 2, y \text{ by } 0.5), \quad B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (\text{rotate } 90^\circ).$$

Compose $T = T_2 \circ T_1: T(x) = B(Ax) = (BA)x$.

- Compute BA :

$$BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix} = \begin{bmatrix} 0 & -0.5 \\ 2 & 0 \end{bmatrix}.$$

- Compute AB :

$$AB = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 0.5 & 0 \end{bmatrix}.$$

Apply both to $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

- $BAx = \begin{bmatrix} -0.5 \\ 2 \end{bmatrix}$ (scale then rotate)
- $ABx = \begin{bmatrix} -2 \\ 0.5 \end{bmatrix}$ (rotate then scale)

They're **different**, so $BA \neq AB$ in general.

(Geometric intuition: doing the spin before the stretch is not the same as stretching before spinning.)

Invertibility of Linear Transformations

- A transformation $T(\mathbf{x}) = A\mathbf{x}$ is invertible iff A is invertible (i.e., $\det(A) \neq 0$)
- Inverse transformation: $T^{-1}(\mathbf{y}) = A^{-1}\mathbf{y}$
- Invertible examples:
 - Identity transformation: $A = I$
 - Rotation by non-zero angle (excluding multiples of 180°)
 - Scaling by a non-zero scalar (e.g., $A = kI$, $k \neq 0$)
 - Any full-rank square matrix
- Non-invertible examples:
 - Projections (lose information)
 - Transformations with zero rows/columns
 - Matrices with linearly dependent columns

Invertibility of Linear Transformations

Invertible (determinant $\neq 0$)

- Identity $A = I$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \det(A) = 1.$$

$$A^{-1} = I.$$

- Rotation (any angle θ , including 180°)

$$R(45^\circ) = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}, \det = 1, R^{-1} = R^\top.$$

$$R(180^\circ) = -I = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \det = 1, \text{inverse is also } -I.$$

- Scaling by a non-zero scalar

$$A = 3I = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}, \det = 9.$$

$$A^{-1} = \frac{1}{3}I.$$

- Any full-rank square matrix (e.g., a shear)

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \det = 1.$$

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \text{ (check } A^{-1}A = I).$$

Inverse transformation demo:

$$\text{Let } A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \text{ and } y = \begin{bmatrix} 9 \\ 0 \end{bmatrix}.$$

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} \Rightarrow$$

$$x = A^{-1}y = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 9 \\ 0 \end{bmatrix} = \begin{bmatrix} 4.5 \\ 0 \end{bmatrix}.$$

$$\text{Indeed, } Ax = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 9 \\ 0 \end{bmatrix} = y.$$

Invertibility of Linear Transformations

Non-invertible (determinant = 0)

- Projection (lose a dimension)

$$P_x = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \det = 0.$$

$[0, 1]^T$ and $[0, 2]^T$ both map to $[0, 0]^T \rightarrow$ can't undo.

- Zero row / zero column

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \text{ (zero row), } \det = 0.$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \text{ (zero column), } \det = 0.$$

- Linearly dependent columns

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \text{ (col}_2 = 2 \cdot \text{col}_1), \det = 0.$$

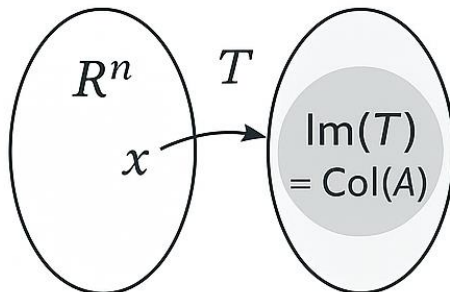
Rank and Image of a Transformation

Let $T(x) = Ax$, $A \in \mathbb{R}^{m \times n}$.

- Image / Range
 - $\text{Im}(T) = \{Ax : x \in \mathbb{R}^n\} = \text{span}\{a_1, \dots, a_n\}$ (the column space).
 - Geometrically: the subspace in $\mathbb{R}^m$ you can reach by A .
- Rank
 - $\text{rank}(A) = \dim(\text{Im}(T))$.
 - Equals the number of linearly independent columns (also = number of non-zero singular values).
 - Bounds: $0 \leq \text{rank}(A) \leq \min(m, n)$.
- Injective / Surjective (using rank)
 - Full column rank ($\text{rank} = n \leq m$) $\Rightarrow$ injective (one-to-one), $N(A) = \{0\}$.
 - Full row rank ($\text{rank} = m \leq n$) $\Rightarrow$ surjective (onto $\mathbb{R}^m$).
 - Square & full rank ($m = n$, $\text{rank} = n$) $\Rightarrow$ bijective = invertible.

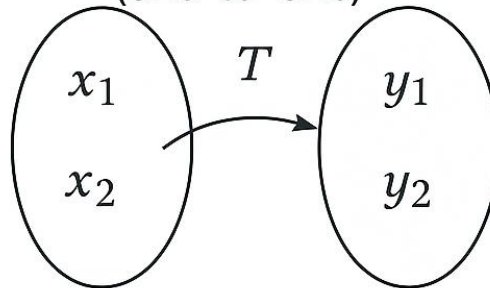
Rank and Image of a Transformation

Image (Range)



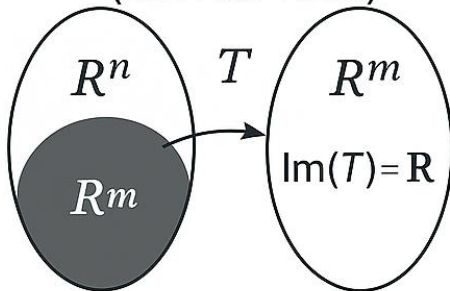
$$\text{rank}(A) = \dim(\text{Im}(T))$$

Injective
(one-to-one)



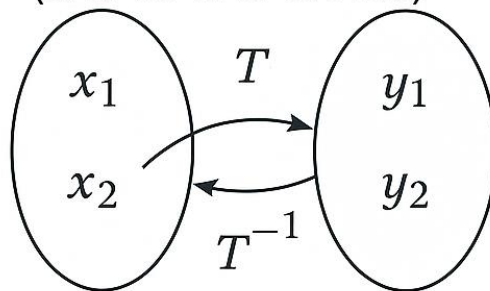
$$\text{rank}(A) = n \leq m$$

Surjective (onto)
(full row rank)



$$\text{rank}(A) = m \leq n$$

Bijjective
(one-to-one & onto)



$$m = n, \text{rank}(A) = n$$

(full column & row rank)

Practice Problems

1. Determine if $T(\mathbf{x}) = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \mathbf{x}$ is linear
2. Construct the matrix for a 90-degree counter-clockwise rotation
3. Show that the projection onto x-axis is a linear transformation
4. Compose a reflection over x-axis with a scaling by 2, and find the resulting matrix
5. Check if $\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$ is invertible
6. Compute the rank of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$
7. Apply $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ to $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and interpret geometrically

Answers to Practice Problems

1. Linearity of $T(x) = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} x$.

Matrix maps are linear: for any α, β and x, y ,

$$T(\alpha x + \beta y) = A(\alpha x + \beta y) = \alpha Ax + \beta Ay = \alpha T(x) + \beta T(y).$$

So T is linear.

2. 90° counter-clockwise rotation matrix (in $\mathbb{R}^2$)

$$R_{90} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

3. Projection onto the x -axis is linear.

Mapping: $(x, y) \mapsto (x, 0)$. Matrix:

$$P_x = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Check: $P_x(\alpha u + \beta v) = \alpha P_x u + \beta P_x v$. Hence linear.

4. Compose reflection over x -axis with scaling by 2.

Reflection: $R = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, scaling (uniform) by 2: $S = 2I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$.

Either order (they commute) gives

$$SR = RS = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

Answers to Practice Problems

5. Invertibility of $\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$.

Determinant $= 2 \cdot 0 - 1 \cdot 0 = 0 \Rightarrow$ not invertible (rank $= 1$).

6. Rank of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$.

Row 2 $= 2 \times$ Row 1 $\Rightarrow$ only one independent row/column $\Rightarrow$ rank $= 1$.

7. Apply $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ to $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and interpret.

$$Ax = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Interpretation: this matrix is a 90° CCW rotation, so the unit vector along x -axis maps to the unit vector along y -axis.

Closing Remarks

- Linear transformations provide a unified view of many operations in AI
- Matrix representation enables efficient implementation and geometric interpretation
- Understanding transformation properties (invertibility, rank) is crucial in ML models